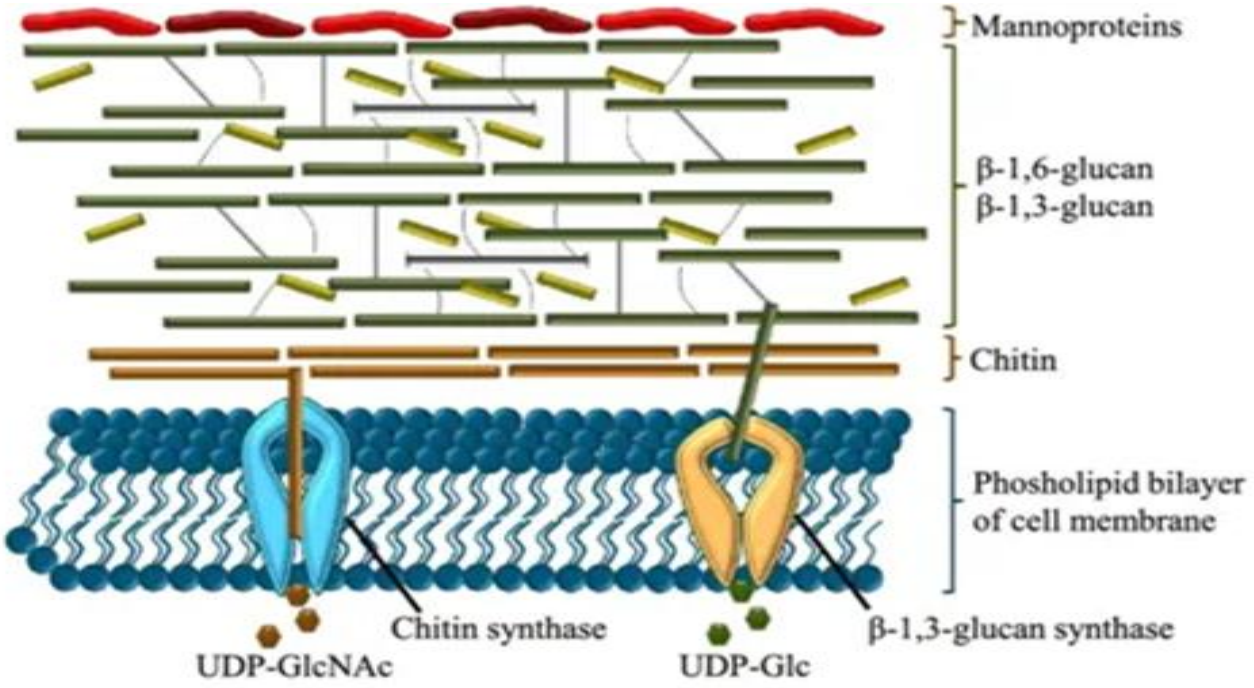
Antifungal drugs

Dr. Begard O. Berzinji PhD Pharmacology

- Fungi, like mammalian cells but unlike bacteria, are eukaryotic and possess nuclei, mitochondria and cell membranes. However, fungi have both a rigid cell wall, primarily made of chitin and glucans, and an inner plasma membrane composed of a phospholipid bilayer with ergosterol



- Many fungal infections occur in poorly vascularized tissues or avascular structures such as the superficial layer of the skin, nails, and hair. Fungi are slow growing and are, therefore, more difficult to kill than bacteria, where cell division can be a target

ANTIFUNGALS

Antifungals often interfere with lipid biosynthesis and disturb the integrity of the fungal membrane (Fig. 1).

A variety of antifungals are used to treat fungal infections, depending on whether the infection is cutaneous or systemic.

Major antifungals include

the polyenes (amphotericin B, nystatin) and

the “azoles” (voriconazole, ketoconazole, and fluconazole).

Antifungal drugs for systemic mycosis

A. Amphotericin B

Amphotericin B is a naturally occurring polyene antifungal produced by Streptomyces nodosus. It has wide spectrum of antifungal activity, and its fungicidal

Mechanism of action: It binds to ergosterol in the plasma membranes of sensitive fungal cells and forming pores (channels)(Figure: 2).

The pores disrupt membrane function, allowing electrolytes (particularly potassium) and small molecules to leak from the cell, resulting in cell death

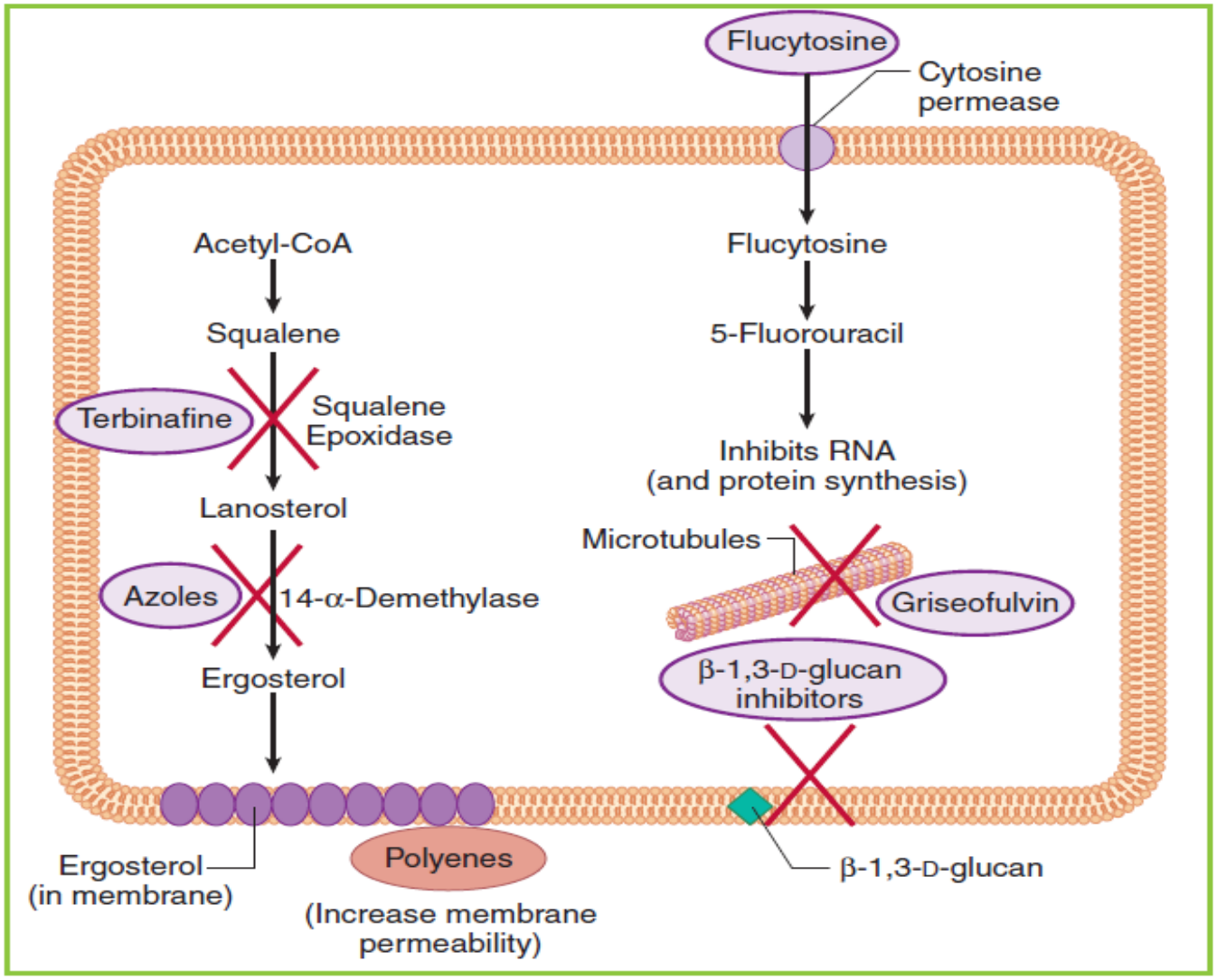
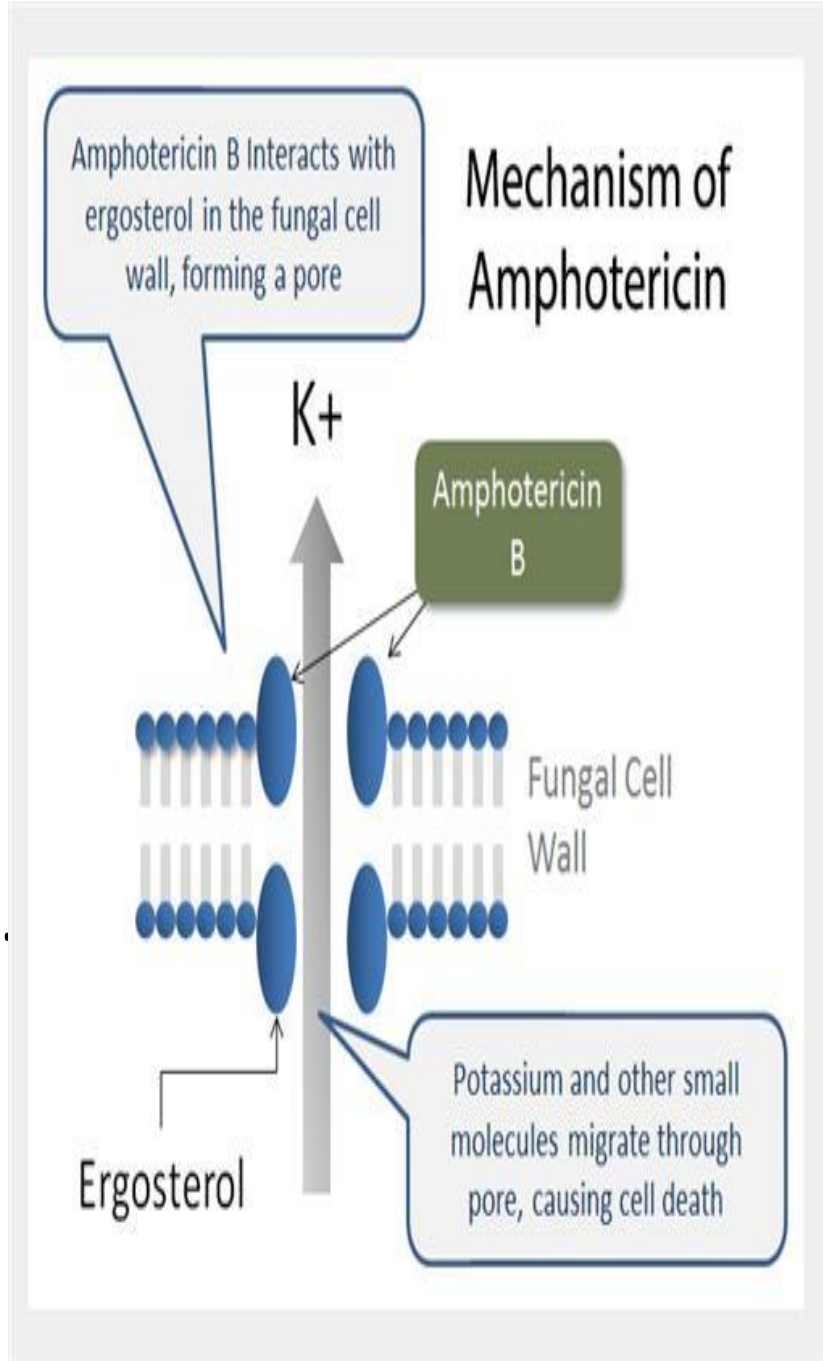

Flucytosine
Cytosine permease
Flucytosine
5-Fluorouracil
Inhibits RNA (and protein synthesis)
Acetyl-CoA
Squalene
Terbinafine
Squalene Epoxidase
Lanosterol
Azoles
14-α-Demethylase
Ergosterol
Microtubules
Griseofulvin
β-1,3-D-glucan inhibitors
Ergosterol (in membrane)
Polyenes
(Increase membrane permeability)
β-1,3-D-glucan

Figure: 1

- Available for topical treatment of common mucocutaneous fungal infections.
- Amphotericin B remains the drug of choice in systemic life-threatening mycosis
- Used IV for deep-seated and severe fungal infections (e.g. Aspergillosis or histoplasmosis).



Adverse-effects: it has narrow therapeutic index

- Fever, chills, headache, nausea, vomiting, and hypotension during intravenous infusion, most commonly 1 to 3 hours after starting the IV administration
- Reversible nephrotoxicity, if detected early. This is dose dependent and almost invariable.

systemic toxicity (especially nephrotoxicity) can be reduced by using the **liposomal/ lipid/micellar formulation**s but they are more expensive

- Electrolyte abnormalities Hypokalemia (requiring potassium supplementation) and hypomagnesaemia;.
- Transient bone marrow suppression.

B. Flucytosin

Its used to treat systemic candidiasis and cryptococcosis,

Mechanism of action: In fungal cell; converted to 5-fluorouracil which disrupt nucleic acid and protein synthesis.(Fig. 3)

- acquired resistance is a major problem therefore, it is only used in combination with amphotericin B in the treatment of cryptococcal meningitis (it penetrates the CSF well) & systemic mycosis.
- Amphotericin B increases cell permeability, allowing more *5-FC* to penetrate the cell leading to synergistic effects

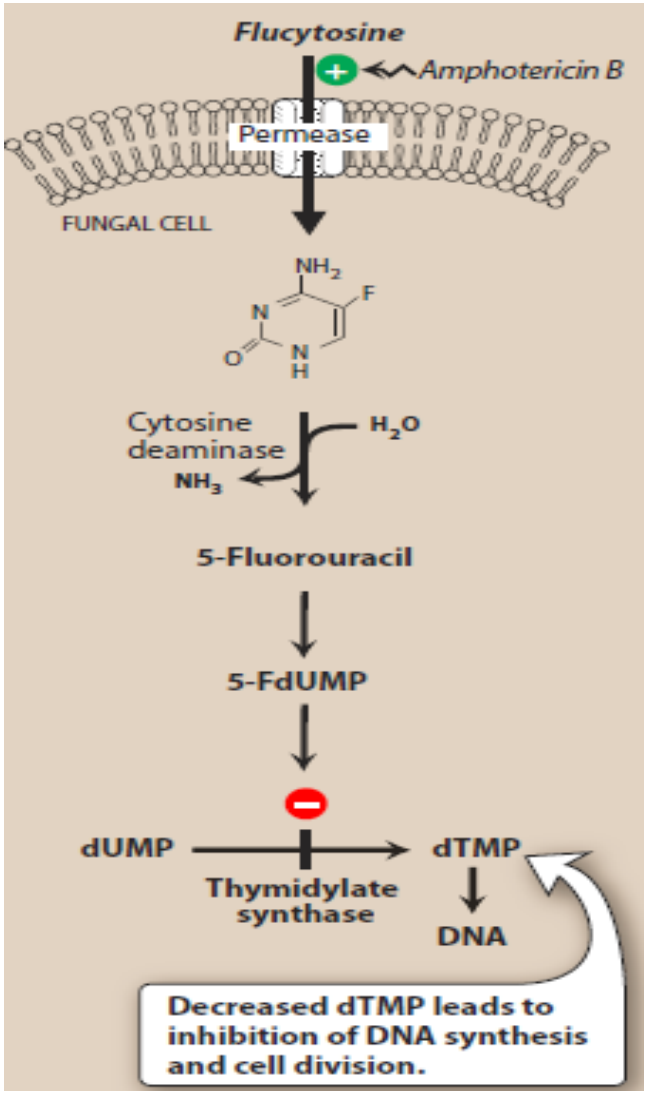


Figure 3

Adverse -effects

- **Bone marrow suppression**: reversible neutropenia, thrombocytopenia
- **Hepatic Dysfunction**: Reversible
- **GI disturbances** (nausea, vomiting, and diarrhea) and severe enterocolitis

C. Azole antifungals

- Azole antifungals are made up of two different classes of drugs—**imidazoles** and **triazoles.** imidazoles are given topically for cutaneous infections, where as triazoles are given systemically for the treatment or prophylaxis of cutaneous and systemic fungal infections.

Imidazole antifungals

Clotrimazole, miconazole, ketoconazole

- Broad-spectrum fungistatic, imidazole drugs used in the topical treatment of oral, skin, and vaginal infections with *C. albicans*

Triazoles: *fluconazole*, *itraconazole*, *posaconazole*, and *voriconazole*

- Given systemically for the treatment or prophylaxis of cutaneous and systemic fungal infections.

Mechanism of action of azoles: act by inhibiting ergosterol synthesis in fungal cell membrane

All azoles inhibit the hepatic CYP450 3A4 isoenzyme, leading to many drug interactions

Considered teratogenic, should be avoided in pregnancy.

Fluconazole

- It is the least active of all triazoles, given orally, IV
- It is active against many *Candida* species, *Cryptococcus neoformans* and *Histoplasma capsulatum*, but it has no role in the treatment of aspergillosis
- also effective against most forms of mucocutaneous candidiasis; thrush, esophageal candidiasis in AIDS patient
- commonly used as a single-dose oral treatment for vulvovaginal candidiasis
- well absorbed, distributes widely to body fluids and tissues. The majority of the drug is excreted unchanged via the urine
- Adverse effects are nausea, vomiting, headache, and skin rashes. Hepatotoxicity can also occur

Itraconazole

- Is a synthetic triazole that has a broad antifungal spectrum compared to *fluconazole*.
- Drug of choice for all forms of blastomycosis, sporotrichosis and histoplasmosis except meningitis,
- Rarely used for treatment of infections due to Candida and Aspergillus species

Voriconazole

- Replaced *amphotericin B* as the drug of choice for invasive aspergillosis. also approved for treatment of invasive candidiasis

D. Echinocandins

Caspofungin, micafungin* and *anidulafungin:

- Have potent activity against Aspergillus and most Candida species, including those species resistant to azoles.

Mechanism of action: interfere with the synthesis of the fungal cell wall by inhibiting the synthesis of glucan, leading to lysis and cell death.

- Most common adverse effects are fever, rash, nausea, and phlebitis at the infusion site. They can also cause a histamine-like reaction (flushing) when infused too rapidly.

Caspofungin

- First-line option for patients with invasive candidiasis
- 2nd-line for invasive aspergillosis in patients who have failed or cannot tolerate amphotericin B or an azole.
- Adverse effects: facial flushing, rash, fever, and pruritus. Dose reductions are required in the presence of moderate hepatic insufficiency

Micafungin and anidulafungin

- They are newer members of the echinocandin class of antifungal drugs. are first-line options for the treatment of invasive candidiasis, including candidemia
- *Micafungin* and *anidulafungin* do not need to be adjusted in renal impairment or mild to moderate hepatic dysfunction

Drugs for cutaneous mycotic infections

A. Squalene epoxidase inhibitors

These agents act by inhibiting squalene epoxidase, thereby blocking the biosynthesis of ergosterol. Accumulation of toxic amounts of squalene results in increased membrane permeability and death of the fungal cell.

1. Terbinafine: Drug of choice for treating dermatophyte onychomycoses (fungal infections of nails).

- Oral *terbinafine* may also be used for tinea capitis (infection of the scalp).
- Topical *terbinafine* (1% cream, gel or solution) is used to treat tinea pedis, tinea corporis (ringworm), and tinea cruris (infection of the groin). Duration of treatment is usually 1 week.

2. Naftifine: Is available for topical use only in the treatment of cutaneous dermatophyte and Candida infections.

B. Griseofulvin

- **Mechanism of action :** causes disruption of the mitotic spindle and inhibition of fungal mitosis.
- Its Fungistatic and requires a long duration of treatment (6 to 12 months for onychomycosis)
- absorbed adequately from GIT and absorption is enhanced by high-fat meals. concentrates in skin, hair, nails & adipose tissue.
- induces hepatic CYP450 activity, which increases the rate of metabolism of anticoagulants. contraindicated in pregnancy and patients with porphyria.

C. Nystatin

Mechanismof action: similar to amphotericin B.

- **Indications:** It is used for treatment of cutaneous mucocutaneous and intestinal infections, especially those caused by Candidal infections.
- Topically for treatment of superficial infections caused by C. albicans
- Orally for the treatment of oropharyngeal candidiasis (oral thrush)
- Intra-vaginally for vulvovaginal candidiasis

D. Imidazoles

- Imidazoles are azole derivatives, they have a wide range of activity
- The topical imidazoles have a variety of uses, including tinea corporis, tinea cruris, tinea pedis, and oropharyngeal and vulvovaginal candidiasis.
- Oral *ketoconazole* has historically been used for the treatment of systemic fungal infections but is rarely used today due to the risk for severe liver injury, adrenal insufficiency, and adverse drug interactions.

E. Ciclopirox

- inhibits the transport of essential elements in the fungal cell, disrupting the synthesis of DNA, RNA, and proteins.

F. Tolnaftate

- distorts the hyphae and stunts mycelial growth in susceptible fungi.

Indication: Both are used for topical treatment of dermatophyte infections and tinea.